Study design — how much skill is there in Champions Reg M-B Bo1?

Drafted 2026-07-25. Status: designed, not run.

This replaces the ratings half of `docs/predictability-study.md`, whose headline — *"the higher-rated player wins 55.0%"* — does not survive the quality filter and, separately, is not the right statistic.

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1. Why the current number is wrong twice

It was measured with bots in

`engine/predictability.py` line 37 excludes a game only if `p1.bot` or `p2.bot` — the **name** flag. That is the rule which missed six high-volume accounts. Measured on the current store:

Filtering	Higher-rated player wins	n
Name flag only — **the published method**	**54.7%** [53.0, 56.5]	3,031
+ behavioural bots removed	**52.4%** [49.9, 54.9]	1,501
Full quality filter	**50.9%** [47.1, 54.6]	676

The middle row is the fair test: it removes only the contamination and changes nothing else. The interval then **includes 50%**. The published claim that rating beats a coin does not survive.

Segmented by ladder tier (bots removed, n=1,501), the "low ladder is the wild west" hypothesis also fails — there is no trend, and the 1200–1300 band is *below* even:

Tier (lower player)	Higher-rated wins
<1100	51.2%
1100–1200	57.3%
1200–1300	46.4%
1300–1400	53.7%
1400+	52.3%

On fully clean games the 1300+ slice is 49.4%.

It is the wrong statistic even when measured correctly

"How often does the higher-rated player win" is a **base-rate statistic about the rating**, not a measurement of skill. It confounds three things: how much true skill varies, how accurately the rating estimates it, and how the observed pairs were selected. `docs/THESIS-REVIEW-v2.md` §F already said this — *"matching two aggregate win rates is not validation (base-rate fallacy)... redo the study as a proper skill/luck decomposition with intervals, à la Lopez et al."* That fix was never applied.

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2. Is the chess comparison fair? No.

Five reasons, in order of severity.

2.1 The ratings are provisional by the platform's own standard. Showdown's Glicko-1 starts every player at R=1500, **RD=130**, and Showdown *refuses to publish a GXE for anyone with RD ≥ 100* because it is "too inaccurate to provide meaningful insight." This store is **three days** of a brand-new format. Essentially every account in it is inside the window Showdown itself considers too uncertain to report. We are not measuring whether skill predicts; we are measuring whether a deliberately-unreliable provisional estimate predicts.

2.2 A Showdown Elo point is not a chess Elo point. Showdown's Elo carries a sliding K-factor and its own decay. A "100-point gap" is not the same unit as 100 FIDE Elo, so mapping it onto chess's expectancy curve is a category error.

2.3 Chess has draws. Chess expectancy is an **expected score** with draws at 0.5. Champions Bo1 has no draws, so 64% expected score and 52% win rate are different quantities.

2.4 Bo1 versus a series. Lopez et al. report the NBA's better team winning roughly 80% of a **seven-game series**; single games are far lower. Real VGC is Bo3. Comparing our Bo1 number to a series number overstates the gap.

2.5 Selection. Lopez et al. use complete league schedules. Public Showdown replays are **opt-in uploads** — players save wins and flashy games. The sample is self-selected in a direction nobody has measured.

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3. What the literature actually offers

Lopez, Matthews & Baumer (2018), *How often does the best team win? A unified approach to understanding randomness in North American sport*, Annals of Applied Statistics 12(4).

This paper exists for precisely our problem: making randomness comparable **across** competitions whose scoring, schedule length and rating conventions differ. Method: Bayesian state-space models fitted to betting-market data, yielding per-league estimates of the **dispersion of team strength**, its between-season / within-season / game-to-game variability, and home advantage.

Reported qualitative findings: the **NBA** has the largest talent dispersion and the largest home advantage; the **NHL and MLB** stand out for the randomness of single-game outcomes; in the NBA the better team wins about **80% of a seven-game series**.

The transferable idea is the comparable statistic. Do not compare raw win rates. Estimate the **standard deviation of latent competitor strength on the log-odds scale (σ)**, then derive **P(better competitor wins a single game)** for a randomly drawn pair. That quantity is defined identically in every sport, and it is what our study should report.

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4. The study

Question. What fraction of a Champions Reg M-B Bo1 outcome is attributable to player skill, and where does that sit against other rated competitions on a common scale?

4.1 Population

Clean games only, funnel reported. Sensitivity runs on (a) bots-removed-only and (b) full filter, because the full filter drops forfeits and short games and therefore conditions on game length.

4.2 Model

Bradley–Terry on **player identity**, not rating:

> $P(i \text{ beats } j) = \sigma(\theta_i - \theta_j), \theta \sim Normal(0, \sigma_\theta^2)$

fitted hierarchically (partial pooling), so a player with four games is shrunk toward the mean instead of posting a spurious extreme. **σ_θ is the estimand** — the dispersion of true skill.

Fitting the *rating* instead would re-import the RD problem from §2.1. Player identity sidesteps it: we estimate skill from results directly, and the platform rating becomes a covariate to validate against rather than the measurement itself.

4.3 The comparable statistic

From the fitted σ_θ, compute by simulation over random pairs:

- **P(better player wins a single game)** — directly comparable to Lopez et al.
- **P(better player wins a Bo3)** — the format real tournaments use.
- **The pregame ceiling:** E[max(p, 1−p)], the best accuracy *any* pregame model could achieve. If that lands near 52%, it explains every null in this project at once — JOLTEON, roles, CHOMP-EV and WAR are not weak models, they are at the ceiling.

4.4 Controls

- Restrict to players with ≥ N games (N = 5, 10, 20) and report σ_θ at each; if σ_θ rises with N, the low-N estimate was noise-dominated.
- Report by rating tier to re-test the "high ladder is more predictable" hypothesis inside the model.
- Bootstrap CIs by **resampling players**, not games — games within a player are correlated.

4.5 Falsification, before trusting the result

Simulate ladders at known σ_θ (0.0, 0.25, 0.5, 1.0) with this store's exact games-per-player distribution, and confirm the estimator recovers each. **An estimator that cannot recover a known σ on this sample size cannot be trusted to report a small one.** Run before touching real data.

4.6 What would falsify the headline claim

σ_θ significantly greater than zero with the interval excluding it, *and* an implied P(better player wins) meaningfully above 0.5 — say ≥ 0.55 — on clean games. Anything less and the honest statement is that pregame skill signal is not detectable at this sample size.

4.7 Honest limits, stated in advance

- Three days of data. No temporal component; σ_θ is a snapshot.
- Opt-in replay upload. Unmeasured selection.
- Small n after filtering (676 fully clean rated games) means **a null here is "not detectable at this sample size", never "proven absent"**.
- Bot detection is a floor. Undetected bots inflate apparent skill dispersion, as demonstrated above.

5. Other rated competitions worth citing

For the comparison table, on the σ / P(better wins) scale rather than raw win rates:

Domain	Why it belongs	Note
Chess (FIDE)	The reference rating system	Must use expected score, draws at 0.5
NBA / NFL / MLB / NHL	Lopez et al., already on a common scale	NBA most predictable, NHL/MLB most random
Tennis (ATP)	Individual, no draws, best-of-N — the closest structural analogue	Bo3 vs Bo5 is a built-in experiment on series length
Esports with public ladders (LoL, CS, SC2)	Same Glicko/Elo family, same opt-in-replay problem	Closest to our data-generating process
Poker	The canonical high-variance skill game	Skill is only detectable over very large samples — the direct precedent for a null here

Tennis and poker are the two most useful. Tennis because a Bo3/Bo5 contrast quantifies exactly what series length buys, which is the question for VGC. Poker because it is the established case where real skill exists and is **undetectable in small samples** — which may be precisely our situation.

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6. Deliverable

`engine/skill_variance.py` → `data/skill-variance.json`, carrying σ_θ with CI, the derived single-game and Bo3 probabilities, the implied pregame ceiling, the simulation recovery check, and the funnel of games it was computed on. Then `docs/predictability-study.md` is rewritten against it, with the superseded 55% figure kept and marked rather than deleted.

Sources

- Lopez, Matthews & Baumer, *How often does the best team win?* — arXiv:1701.05976 · Annals of Applied Statistics
- Pokémon Showdown ladder help — Elo with sliding K-factor and decay
- Glicko rating system — RD and its meaning
- Everything You Ever Wanted to Know About Ratings, Smogon — GXE withheld at RD ≥ 100